PROBLEM FORMULATION, ENVIRONMENT CLASSIFICATION & STATE TYPE

WATERTANK: SMART WATER TANK CONTROLLER AI AGENT

1. Problem Formulation

WaterTank evaluates the tank, source and motor conditions to recommend a safe motor action for the current time. Its decisions must respect the dry-run limit, overflow limit, motor health and power availability. If no safe action can meet the water-level goal, the agent alerts the user instead of forcing the motor to run.

Element	Description
Initial State	The agent receives the tank level, sump level, motor status and health, power availability, usage history, and the safe limits (for example LOW = 25%, HIGH = 90%).
Actions	Update sensor information; generate possible actions; check safety constraints; turn the motor ON; turn the motor OFF; keep the current state; and alert the user when no safe action exists.
Transition Model	When water is used, the motor runs, power fails, or a fault occurs, the tank-state model is updated. Motor ON raises the tank level, motor OFF stops filling, and usage lowers the level, so the set of safe actions may change.
Path Cost	Among safe actions, reduce motor starts/stops, electricity use, and time spent below the comfort level. Hard constraints (no dry running, no overflow, no faulty motor operation) cannot be traded away to reduce cost.
Goal Test	The tank level stays between LOW and HIGH without violating any safety limit. If this is not possible, the case is explicitly flagged and the user is alerted.
State Space	All possible tank conditions represented by tank level, sump level, motor status and health, power status, time, usage pattern, and pending alerts.
Search Space	Possible sequences of reading sensors, updating the model, checking constraints, comparing safe actions, acting or alerting, and rechecking after new events.
Agents	WaterTank is the intelligent decision-support agent. Household users and the power supply affect the environment, but WaterTank does not repair equipment or override manual control.

2. Environmental Properties

Property	Classification	Reason
Observability	Partially Observable	The agent depends on sensor data. Readings may be noisy or wrong, and leaks or blockages may not be detected immediately.
Determinism	Stochastic / non-deterministic	Water usage, power cuts, and motor problems can change the expected water level.
Episodes	Sequential	Running the motor now affects the water level and the sump level for later decisions.
Change	Dynamic	Tank level, sump level, and power availability change while the agent is operating.
Values	Discrete & Continuous	Motor ON/OFF and alerts are discrete, while water level, time, and usage are numerical.
Agents	Single-agent environment	WaterTank is the only decision-maker. Household water use and power cuts are treated as environment events, not competing agents.

3. Choosing the State Type

The state type describes how the agent represents the situation it is solving. WaterTank must maintain a model of changing tank and motor conditions, compare possible actions, and revise its decision when new sensor data arrives.

Option	Suitability	Explanation
Single-state	Not suitable	A single snapshot cannot represent the changing water level, motor status, power supply, and usage that affect the next decision.
Multi-state	Most suitable	The agent moves among many possible tank states: current level and motor status, candidate actions, updated levels after filling or usage, and alert states requiring user action.

CONCLUSION

WaterTank is best represented as a multi-state problem because it evaluates many possible combinations of tank level, motor condition, power availability, and actions. It is also partially observable and stochastic: the agent relies on available sensor data, while unexpected usage, power cuts, or faults can change the situation. Its planning is sequential because each motor action affects the later water level.